2026-1 Social Intelligence

Final Project Introduction

- Sentiment Analysis Modeling and Becoming an E-Commerce Data Scientist -

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Final Project: Social Review Data Analysis

Why This Project — Review Data Is a Goldmine

Social Review Data — the richest and noisiest business data on the planet

Every time you leave a star rating and a one-line review on **Coupang, Amazon, or Naver Shopping**, that data is being collected somewhere.

This data is —

- **Massive in scale** — thousands to millions per product
- **Free-form** — text, ratings, photos, videos
- **Unlabeled** — ground truth is unclear or missing
- **Noisy** — spam, sarcasm, jokes, typos

→ In short, it's the **exact opposite of textbook datasets**. This is what you'll actually face in practice.

Examples of Review Data

리뷰 742

가벼워요 | 휴대하기 편해요 | 성능이 좋아요 | 색상이 예뻐요 | 속도가 빨라요 | 배터리가 오래가요 | 디자인이 예뻐요 | 사용이 편해요 | 크기가 적당해요
이동이 편리해요 | 편해요 | 작업하기 좋아요 | 사이즈가 작아요 | 충전이 편해요 | 키감이 괜찮아요 | 쓰기 좋아요 | 수납이 잘 돼요 | +3

전체 | 상품옵션 — 랭킹순 · 최신순 · 평점 높은순 · 평점 낮은순

리뷰 클렌징 시스템 작동 중입니다.

★ 5 · BEST
pcsi****** · 26.02.05. · 신고

제품 선택: 14ZD95U-GX56K/R5/램16 / SSD 256GB 무상지원: 노트북 개봉 후 SSD256 장착 +0 / 윈도우11패키지 할인 이벤트: 윈도우11 할인 개봉설치 +9.9만

무게 가벼워요 | 성능 좋아요

대학교 입학을 앞두고 노트북을 알아보다가, 이미 대학생활을 하고 있는 사촌언니 추천으로 LG 그램을 구매하게 됐어요...!

1. 무게와 휴대성
그램을 처음 들었을 때 제일 먼저 든 생각은 진짜 가볍다였어요. 한 손으로도 쉽게 들릴 정도라서, 전공책만 넣어도 무거워지는 가방에 넣어도 부담이 없어요. 이동이 많은 대학생에게는 이 부분이 정말 큰 장점이라고 느꼈어요! ...
더보기

리뷰가 도움이 되었나요? 도움돼요 7

상품 리뷰

★★★★★ 71

최고		90%
		8%
		0%
		1%
		1%

있어요 91%

자세히 보기

대해 작성된 상품 리뷰로, 판매자는 다를 수 있습니다.
상품리뷰 운영원칙

베스트순 최신순 | 검색어를 입력하세요 | 모든 별점

안우상 실명리뷰어
★★★★★ 2026.03.20
판매자: 쿠팡(주)

오뚜기 식이섬유 플러스 현미밥, 150g, 30개

오뚜기 식이섬유 플러스 현미밥 내용 리뷰 재구매 의사

오뚜기 식이섬유 플러스 현미밥 내용 리뷰
재구매 의사
간편한 건강식으로 O 전자레인지 돌리면 바로 먹을 수 있어서 바쁜 날 진짜 유용해요 ㅎㅎ 건강 챙기면서 간편함까지 잡은 느낌이라 계속 사게 될 듯해요
다이어트 식단용으로 O 일반 흰밥보다 부담 적고 포만감도 괜찮아서 식단 관리할 때 은근 도움돼요 ㅋㅋ 꾸준히 먹기 괜찮아요
집에 쟁여두는 용도로 O 130g이라 양 조절하기 좋고 여러 개 쟁여놓으면 한 끼 해결 편해서 만족이에요 ^^
장점
일단 식이섬유가 한 개당 6g 들어 있어서 일반 즉석밥보다 확실히 건강 느낌이 있어요 �
하루 권장량의 약 24퍼센트 수준이라 한 끼만 먹어도 꽤 채워지는 점이 장점이에요 �
현미랑 보리 섞인 식감이라 씹는 맛이 살아있고, 흰밥보다 훨씬 고소해요 ㅎㅎ
130g 소용량이라 과식 안 하게 되는 것도 좋고, 다이어트나 식단 관리할 때 딱 좋아요
전자레인지 1~2분이면 끝이라 진짜 편해요 ㅋㅋ 바쁠 때 최고
G-Enews
Viva 100
단점
현미 특유의 거친 식감 때문에 부드러운 밥 좋아하는 사람은 약간 불호일 수 있어요
양이 적어서 일반 성인 기준으로는 반찬 없이 먹으면 살짝 부족하게 느껴져요 ㅠㅠ
가격이 일반 즉석밥보다 조금 더 있는 편이라 가성비만 보면 고민될 수 있음
잡곡 느낌이 강해서 호불호는 확실히 갈릴 수 있어요
총평
이건 그냥 "건강 챙기는 즉석밥"이라고 보면 딱 맞아요 ㅎㅎ
식이섬유 함량이 높은 게 확실한 차별점이라 다이어트나 장 건강 신경 쓰는 분들한테 특히 괜찮은 제품이에요
맛도 생각보다 고소해서 계속 먹기 나쁘지 않고, 간편함까지 있어서 바쁜 일상용으로 잘 맞아요 ㅋㅋ
다만 양이 적고 식감이 호불호 있다는 점만 감안하면, 건강식 즉석밥으로는 충분히 만족도 높은 제품이에요

맛 만족도 맛있어요

16명에게 도움이 됐어요 — 신고하기

What Do Companies Do With This Data?

Real business problems that your project mirrors

🛒 Search & Recommendation

Coupang / Amazon / Netflix

Extract meaning from review text to power semantic search and **personalized recommendations** — capturing intent that simple keyword matching misses.

📈 Product Improvement (VoC)

Samsung / Apple / Tesla

Voice of Customer — which features draw the most compl aints? Directly drives prioritization for next models or OTA updates.

🎯 Marketing & Brand Monitoring

P&G / Coca-Cola

Track sentiment shifts after campaigns, monitor competitive positioning, detect crises early through review-stream anoma lies.

🤖 Generative AI Training

OpenAI / Anthropic

Reviews are raw material for LLM alignment and evaluation — human preference signals feed RLHF and DPO reward models.

Project Overview — Two Tasks, 100 Points Total

Task 1 (30 pts) + Task 2 (70 pts)

This project consists of **two independent tasks**. **Task 1** is a *classification problem* with a defined ground truth, while **Task 2** asks you to *analyze unlabeled e-commerce data* through the lens of a data scientist.

Task 1 — 30 pts

Sentiment Classifier

4-class Sentiment Classification

- **4-class labels (negative ~ positive)**
- Noisy / biased training data
- Metric: **Quadratic Weighted Kappa**

Task 2 — 70 pts

E-Commerce Data Scientist

Retrieval + Insight Discovery

- **2.1 Build a Retriever** (30 pts)
  Retrieval pipeline (MAP)
- **2.2 Discover Insights & Value** (40 pts)
  Data → Insight → Value (Qualitative evaluation)

Build a Sentiment Classifier

Data — Real-World Training Set / Curated Test Set

Two datasets, very different quality — understand the gap

Real-World Data (Training — Provided)

- Some level of label error is expected
- Labels are biased (class imbalance, domain skew)
- Mixed-quality text — typos, noise, slang, …
- You may analyze and re-label as you see fit
- Free choice of noisy-label / cleaning strategy

Test Data (Evaluation — Hidden)

- Held by instructor — not released to students
- Labels refined through LLM-based cleaning
- Higher-quality, more reliable labels
- Final score is computed on this curated set
- Train / test label distributions may differ

Note: **Preprocessing** may improve performance and make experimentation easier.

I recommend splitting the training set into **train/validation** for proper evaluation.

Labels — 4-Class Ordinal

Order matters — a sentiment scale, not just four arbitrary buckets

negative	weak_negative	weak_positive	positive
Strong negative	Weak negative	Weak positive	Strong positive

Increasing positivity →

Why this matters

Confusing **negative** ↔ **positive** is a big mistake. Confusing **weak_negative** ↔ **weak_positive** is a small mistake.

Plain accuracy *treats both equally* — which is wrong.

→ We need a metric that **weights errors by how far apart they are**. That metric is QWK (next slide).

Metric — Quadratic Weighted Kappa (QWK)

Definition — agreement above chance, weighted by class distance

$$\kappa = 1 - \Sigma_{ij}\, W_{ij}\, O_{ij} \,/\, \Sigma_{ij}\, W_{ij}\, E_{ij}, \qquad W_{ij} = (i - j)^2 / (N - 1)^2$$

Notation

- O_{ij} — observed counts (true=i, pred=j)
- E_{ij} — expected counts under chance
- W_{ij} — distance weight (further off = larger)
- **N** — number of classes (= 4)

Range: **−1 to 1** (1 perfect, 0 chance)

Implementation: `quadratic_weighted_kappa` in `utils.py`

Weight Matrix W (4×4)

	neg	w-neg	w-pos	pos
neg	0	1/9	4/9	1
w-neg	1/9	0	1/9	4/9
w-pos	4/9	1/9	0	1/9
pos	1	4/9	1/9	0

Dataset


```json
[
  {
    "review_id": 566259302,
    "product_name": "iphone_15",
    "content": "배송 박스부터 기분이 안좋았으나 와이프 핸드폰을 빨리 바꿔주고싶어 오픈하였습니다.\n\n보시다시피 아이폰15 박스 왼쪽, 오른쪽상단,중간까지
    "label_string": "negative",
    "label": 0
  },
  {
    "review_id": 586222236,
    "product_name": "iphone_15",
    "content": "별점을 1점 주는 이유는 쿠팡에게 주는 거 아니에요\n저는 쿠팡 하루, 이틀 간격으로 꾸준히 구매하는 구매자이고\n냉정하게 아이폰 15한테 주는
    "label_string": "negative",
    "label": 0
  },
  {
    "review_id": 759633379,
    "product_name": "iphone_15",
    "content": "제목 그대로 이게 새상품이 맞나 싶어요..\n기대하며 샀는데 기존에 쓰던 6년 가까이 쓰던 아이폰 11과 다를바가 없어요. 발열이 우선 엄청 심합
    "label_string": "negative",
    "label": 0
  },
  {
    "review_id": 626721840,
    "product_name": "iphone_15",
    "content": "배송이 왔을때부터 걸레짝마냥 다 찌그러져있는 배송상태며\n핸드폰 케이스 상자 또한 그 덕에 찌그러져있었지만\n그래도 새 핸드폰을 받는 다는
    "label_string": "negative",
    "label": 0
  },
  {
    "review_id": 636039366,
    "product_name": "iphone_15",
    "content": "빠른배송과 예쁜디자인 아이가 모두 만족해 하며 좋아했습니다.\n다 너무 좋았는데.....\n빠른배송을 하시느라 핸드폰인데도 9월21일 강원도뿐
    "label_string": "negative",
    "label": 0
  },
  {
    "review_id": 558798630,
    "product_name": "iphone_15",
    "content": "와.. 돈이 130만원이 넘는데..\n배송되자마자 문열고나갔더니\n상자가 손이 쑤욱 들어가고 물건바꿔치기했나싶을정도로 구멍이 크게 뚫려있네요.
    "label_string": "negative",
    "label": 0
  },
```

Become an E-Commerce Data Scientist

Task 2 — Two Parts and the Dataset

First build a retriever, then discover insights using it and other tools

2.1 Build a Retriever (30 pts)

- Retrieval pipeline.
- Query expansion → Retrieval → Reranking.
- Keyword and semantic retrieval supported.

2.2 Discover Insights & Value (40 pts)

- Apply methods (topic modeling, networks, clustering, Information Retrieval, generative AI)
- Discover Insights and translate them into value


```
[
  {
    "product_name": "iphone_17",
    "reviewId": 820835950,
    "productId": 9024167492,
    "vendorItemId": 93437609248,
    "itemId": 26462329867,
    "rating": 1,
    "helpfulCount": 18,
    "helpfulTrueCount": 11,
    "helpfulFalseCount": 7,
    "itemName": "Apple 아이폰 17 자급제, 화이트, 512GB",
    "itemImagePath": "https://thumbnail.coupangcdn.com/thumbnails/remote/640x640/image/retail/images/23578657941381-0529
    "title": "어떻게 배송했길래....박스가 다 구겨지고 뜯겨있음",
    "content": "얼마나 박스를 발로 밟았는지 던졌는지 다 찌그러져 있고 심지어 박스 밀봉도 심하게 뜯겨 있어서 내용물이 다 보이네요\n\n심지어 아이폰 박스도
    "reviewAt": 1766491920000,
    "member": {
      "type": "USER",
      "name": "이*연",
      "email": "yyo***@gmail.com"
    },
    "displayName": "이*연",
    "displayWriterIconUrls": [],
    "reviewSurveyAnswers": [
      {
        "reviewSurveyQuestionId": null,
        "question": "가성비",
        "answer": "성능에 비해 비싸요",
        "questionType": "MULTIPLE_CHOICE"
      },
```


Dataset: **smartphone reviews** (multiple Samsung / Apple models, text + metadata). No ground-truth labels.

Task 2.1 — Offline Preparation

Three things you build before any query comes in

Retrieval Backbone

- **BM25** for keyword / sparse retrieval
- **Pretrained embedder** for dense retrieval
- Both at once (**hybrid**) is fine
- Multiple embedders / ensembles allowed
- **Domain adaptation, fine-tuning (optional)**

Metadata Extraction (optional)

- Use only if it helps your retrieval
- Useful for filtering / boosting
- e.g., product, rating, date, brand
- **LLM-based attribute extraction** is fine
- **Skip entirely if your pipeline doesn't need it**

Self-built Evaluation Set

- Optional but **strongly recommended**
- **Craft your own (query, doc) labels**
- Cover both keyword and semantic queries
- Helps you debug and tune the retriever

Why this preparation matters

Retrieval quality is mostly decided **before you ever issue a query**: what you embed, how you index, and what metadata you have all set the ceiling.

A self-built eval set lets you **iterate fast** instead of guessing.

Task 2.1 — Online Retrieval Pipeline

Three stages — each one is a place to be creative

1 Query Expansion

Synonym lookup, LLM rewrite, multi-query generation, etc

→

2 Single / Parallel Retrieval

Sparse (BM25) + Dense (embe dding) hybrid is fine

→

3 Reranking & Post-processing

Cross-encoder, metadata filters, deduplication, etc

What you can experiment with

- **Different expansion strategies** — does LLM rewriting actually help, or hurt for short keyword queries?
- **Different retrieval mixes** — Any retrieval approach is allowed — e.g., dense, sparse, hybrid, with any weighting or tuning.
- **Different rerankers** — cross-encoder vs. LLM listwise, on top-100 or top-50.

Task 2.1 — Evaluation (LLM-as-Judge → MAP)

Instructor uses an LLM to label relevance, then scores retrieval with MAP

① **Test Query Selection**

Instructor curates ~25 queries — a balanced mix of keyword, semantic, and ambiguous queries

② **LLM-as-Judge Generates Ground-Truth Labels**

An LLM judges (query, document) pairs as relevant (1) or not (0). Fixed prompt / model / seed for full reproducibility

③ **Each Team Returns a Ranked List**

Your retrieval pipeline outputs a ranked list per query

④ **Score with Mean Average Precision (MAP)**

Per-query MAP = Σ (precision@k $\times$ rel_k) / |relevant docs|. Captures both rank position and relevance

Intuition: **the more relevant docs near the top, the higher you score** — rewards both recall and ranking quality

Task 2.2 — How Data Scientists Surface Insights

Combine these methods freely to surface business value

① Topic Modeling

LDA / BERTopic to surface latent themes in reviews

② Network Analysis

Co-occurrence graphs,
Keyword association analysis

③ Clustering

K-means / HDBSCAN on embeddings to group reviews, products

④ Information Retrieval

Use the 2.1 retriever to surface specific themes / complaints

⑤ Generative AI

LLMs for summarization, classification, persona generation

⑥ Anything Else

Statistical tests, time-series analysis, regression, etc.

Evaluation

Grading

Idea 40 · Performance 40 · Code 20

Idea/Novelty · 40 %

- Originality of the question
- Methods and techniques you tried
- Depth of analytical perspective
- Connections you draw across data

Performance/Content · 40 %

- Quantitative score (Task 1: QWK, Task 2.1: MAP)
- Persuasiveness of findings (Task 2.2)
- Sufficiency of evidence
- Business significance / impact

Code/Presentation · 20 %

- Reproducibility, I/O contract, runnability (see next slide)
- Report writing & narrative
- Visualization quality
- Slide / document clarity

Examples of strong insights (not exhaustive)

- Users often share fixes in negative reviews → opportunity for guided onboarding
- Positive reviews focus on real use-cases → shift marketing from specs to scenarios
- Low ratings frequently mention delivery issues → improve shipping reliability and communication

Code (Reproducibility · I/O · Runnability)

Ensure your code can be run end-to-end and produces consistent results when re-run.

①
Fixed Random Seeds

Consider controlling randomness (e.g., torch / numpy / random) to improve reproducibility.

②
Tame Probabilistic LLM Outputs

When using LLMs: temperature = 0 or fixed seed. Pin model versions for external APIs (e.g., gpt-4o-2024-08-06)

③
Consistent Input / Output Contract

Match function signatures exactly. Task 1: predict_sentiment(text:str) -> str must return one of the four labels

④
Runnable Out of the Box

Provide runnable code (e.g., sentiment.py, elastic_search.py). Use uv for environment setup and avoid hard-coded paths.

⑤
Report = Your Reasoning

Focus on the process: hypothesis → experiment → interpretation. Include insights from both successful and unsuccessful attempts where relevant. Use visuals to clearly present your findings.